

Universidad Nacional Autónoma de México

OAN–SPM Catalog of Galactic Stellar Clusters

Doctoral Thesis Candidacy

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Abstract

The study of stellar clusters within the Galaxy serves an important role in understanding the Galaxy's physical and chemical characteristics. Open stellar clusters are defined as gravitationally-bound groups of stars born from a single, homogeneous molecular cloud, typically composed of a single stellar population. This allows us to measure physical parameters such as a cluster's age, metallicity, distance, and extinction, which can be determined through photometric data of resolved cluster members. Studies of Galactic clusters have seen a significant boon in recent years with the advent of Gaia, whose unmatched astrometric precision has allowed for the identification of thousands of new stellar clusters, as well as refining membership lists for previously identified clusters.

This doctoral thesis project makes use of Gaia DR3 (and DR4 in the near future) data to update and expand the membership lists from the Hunt & Reffert (2024) and the Unified Cluster Catalog (UCC). By using HDBSCAN we identify an additional population of faint stars for each cluster, extending beyond the main clusters' cores. We determine each cluster's physical parameters using both isochrone fitting methods such as ASteCA, as well as a neural network trained on observational Hertzsprung-Russell diagrams in various passbands (Gaia, 2MASS, WISE, SDSS), including Johnson-Cousins UBVRI passbands from observations made at the Observatorio Astronómico Universitario – San Pedro Martir. Multi-band observations, as well as the recovery of previously unreported faint ($G > 18$) members, will help us break degeneracies present in the parameter space responsible for parameter discrepancies across catalogs.

1 Introduction

Given that stars make up the fundamental blocks of the Galaxy, stellar clusters (SCs) therefore act as important astrophysical laboratories ideal for understanding It's dynamics and structure. Stars are well known to be born in cohorts within molecular clouds, dissolving into the interstellar medium to form the field stars that can be seen today. From current star formation theory we know that the youngest of these clusters where formed from a singular burst of star formation, and are therefore referred to as *simple stellar populations* (SSPs), groups of chemically homogeneous stars (Huang et al., 2024). These groups are known as *open clusters* (OCs) due to their loosely-bound and irregular appearance. OCs also tend to exhibit lower total stellar masses, and are primarily located in the Galactic disk (Zwart et al., 2010). OCs can be used as tracers for studying the Galactic spiral structure (Dias et al., 2022).

Globular clusters (GCs), on the other hand, are high-density collections of stars, distributed throughout the Galactic halo. GCs are found to be in a more stable virial state, with an older age distribution as well as higher dynamical timescales, which contributes to their spherically symmetrical appearance. As a result of their larger age, GC populations tend to be more complex, to the point that they sometimes cannot be properly described as a SSP; chemical abundance variations between stars in a single GC indicates *multiple stellar populations* (MPs) with later-generation stars showing

distinct chemical compositions (Huang et al., 2024). Although they have historically been studied as a separate category from OCs, current models do not discriminate between either OCs or GCs for describing their dynamics and compositions (Krumholz et al., 2019).

Photometric studies of SCs involve fitting synthetic isochrones to color–magnitude diagrams (CMDs). Each isochrone depends on the given age, metallicity, distance, and reddening parameters; each parameter is simultaneously adjusted such that the dispersion around the resulting isochrone is minimized. Stellar isochrones are generated with codes such as PARSEC (Bressan et al., 2012) for a given grid of parameters, which are then fit through visual analysis of the CMDs (which inevitably introduces a level of bias into the solution) or by automated methods such as ASteCA (Perren et al., 2015) or a cross-entropy optimization algorithm with a likelihood function defined by the distance of each point to the nearest point on the isochrone (Monteiro et al., 2010). Recent efforts have been made in the characterization of open clusters in today's age of Big Data, with machine learning tools being made to characterize each cluster from the observed CMDs (Cavallo et al., 2023; Hunt & Reffert, 2023). Large samples of Galactic SCs enable the morphological and chemical distribution through the Galaxy (Gaia Collaboration, Recio-Blanco, et al., 2023; Hu & Soubiran, 2025).

1.1 Objectives

Although a variety of studies have been done using CMDs from other catalogs, few studies include observations in near-ultraviolet filters due to the demanding photometric conditions (as a result of atmospheric sensitivity) required to obtain reliable measurements for a cluster's metallicity and age. For this project, we will exploit over 20 years of UBVRI observations of over 1700 resolved Galactic clusters (both GCs and OCs), which can be used to provide reliable measurements of the clusters' fundamental parameters with stronger model constraints than is otherwise possible without UBV observations (Kaviraj et al., 2007). The observations from the *Observatorio Astronómico Nacional – San Pedro Mártir* (hereafter referred to as SPM) are complemented with photometry obtained from all-sky surveys when available, crossmatching the Gaia observations with GALEX, 2MASS, PAN-STARRS, SDSS, and WISE.

The project's key objectives are outlined as follows:

- Determine and expand on stellar cluster membership catalogs using high quality astrometric data, focusing on retrieving faint member stars that were otherwise discarded in previous studies.
- Implement existing tools for determining clusters' physical parameters based on cleaned CMDs using revised membership lists.
- Develop new tools for the efficient and accurate characterization of cluster CMDs.
- Use newly developed tools for determining stellar and cluster parameters of SCs and SC candidates using observations from SPM, supplemented with observations from external catalogs: 2MASS, WISE, GALEX, Gaia, PAN-STARRS, and SDSS when available.

2 Catalog of Observations

For this project we have observations of over 1700 clusters, compiling the results from 298 observation campaigns dating back to the year 2001. Each observational campaign focuses either on the suspected cluster members, or the foreground or background stars. Doing a preliminary cross-match with the *Unified Cluster Catalog* (UCC; Perren et al., 2023) shows over 300 clusters not identified in the UCC, with some having been identified as asterisms (physically unrelated groups of observed stars). As part of the initial data exploration step of the methodology, a more in-depth review of the published literature will be needed to determine the characterization already performed for each cluster in the sample.

As can be seen in Figure 1, our observations encompass a dense sample the Solar neighborhood, both spatially as well as characterization quality from the UCC UTI metric, providing a good opportunity to improve the worst-scoring clusters and provide a uniform study of the stellar parameter spatial distributions. Thanks to the photometric conditions of at SPM, stars up to a magnitude of ~ 21 can be recorded, allowing for a deeper high quality measurement of SC members. This will allow us to obtain quality photometric measurements for a higher amount of low luminosity members, thus reducing selection bias effects as a result of incomplete sampling of low mass members.

2.1 Initial Sample

As a first test of the membership determination pipeline and preliminary evaluation of the physical characterization of each OC's CMD, we selected four well-studied, well-populated clusters all located within a similar parallax range.

3 Membership Determination

In order to properly characterize a SC, it's member stars must be properly identified to avoid including field stars not bound to the cluster in question for the ensuing CMD analysis. As a result, cluster membership determination algorithms have received a large amount of attention in recent years. For this project we implement a membership determination pipeline using HDBSCAN, an improvement over the DBSCAN clustering algorithm by assembling minimum spanning trees and subsequently compacting them into a set of "groupings," from which stellar clusters are identified. HDBSCAN has been used in recent works with a high rate of success in both identifying previously unknown SCs as well as expanding on known clusters' membership lists (Hunt & Reffert, 2021; Sánchez-Sanjuán et al., 2024; Tarricq et al., 2022).

3.1 Gaia DR3 Data Selection

Our HDBSCAN pipeline operates on the Gaia DR3 (GDR3) (Gaia Collaboration, Vallenari, et al., 2023) 5D astrometric phase space $(\alpha, \delta, \omega, \mu_\alpha \cos\delta, \mu_\delta)$; radial velocities are not included in order to maximize our sample size, as few stars within the

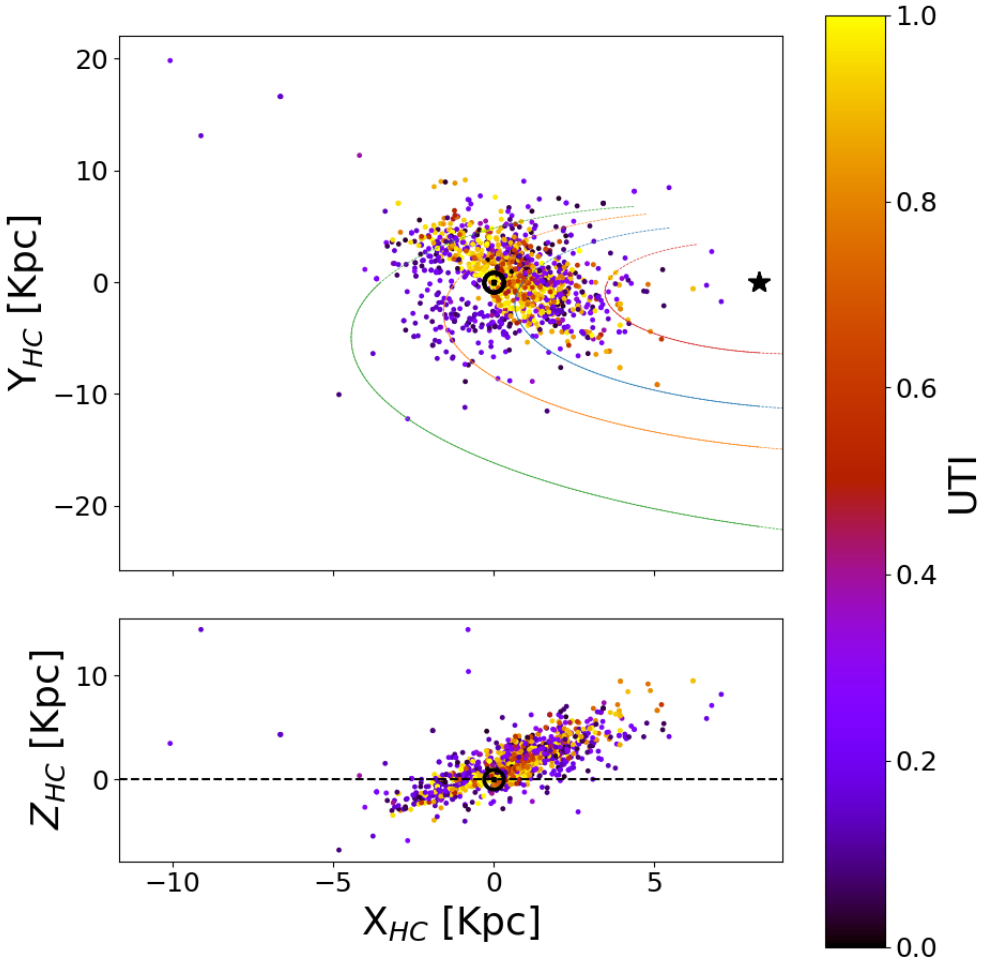


Figure 1: Spatial distribution of observed SCs in the Galaxy in the Solar neighborhood. Heliocentric coordinates were calculated using distances determined through manual photometric analysis of each cluster.

GDR3 catalog include radial velocity measurements. For each of the four clusters analyzed for this work we selected all stars within a conical field centered on the cluster reported center. The search radius was taken as a multiple of the literature reported cluster radius: $\texttt{max}\,(10 \cdot R_{50,\ \mathrm{UCC}},\ 2 \cdot R_{\mathrm{tot},\ \mathrm{HR24}})$, where $R_{50,\ \mathrm{UCC}}$ is the half-member radius reported in the UCC, and $R_{\mathrm{tot},\ \mathrm{HR24}}$ is the tidal radius reported by Hunt & Reffert, 2024.

Each astrometric parameter was normalized to avoid the different scales of each parameter to overpower the other four dimensions, in particular to avoid issues with each field's parallax range, as its scale is order of magnitudes smaller than the other parameters. This was done using the `RobustScaler` scaler from `scikit-learn` (Pedregosa et al., 2011), which removes the data median and scales the data with the quantile range to avoid scaling problems due to outliers. We performed a quality cut on our initial field sample by retrieving all stars in the field that satisfy $\texttt{fidelity_v2} \geq 0.5$, as published and recommended by Rybizki et al., 2022, where the `fidelity_v2` flag

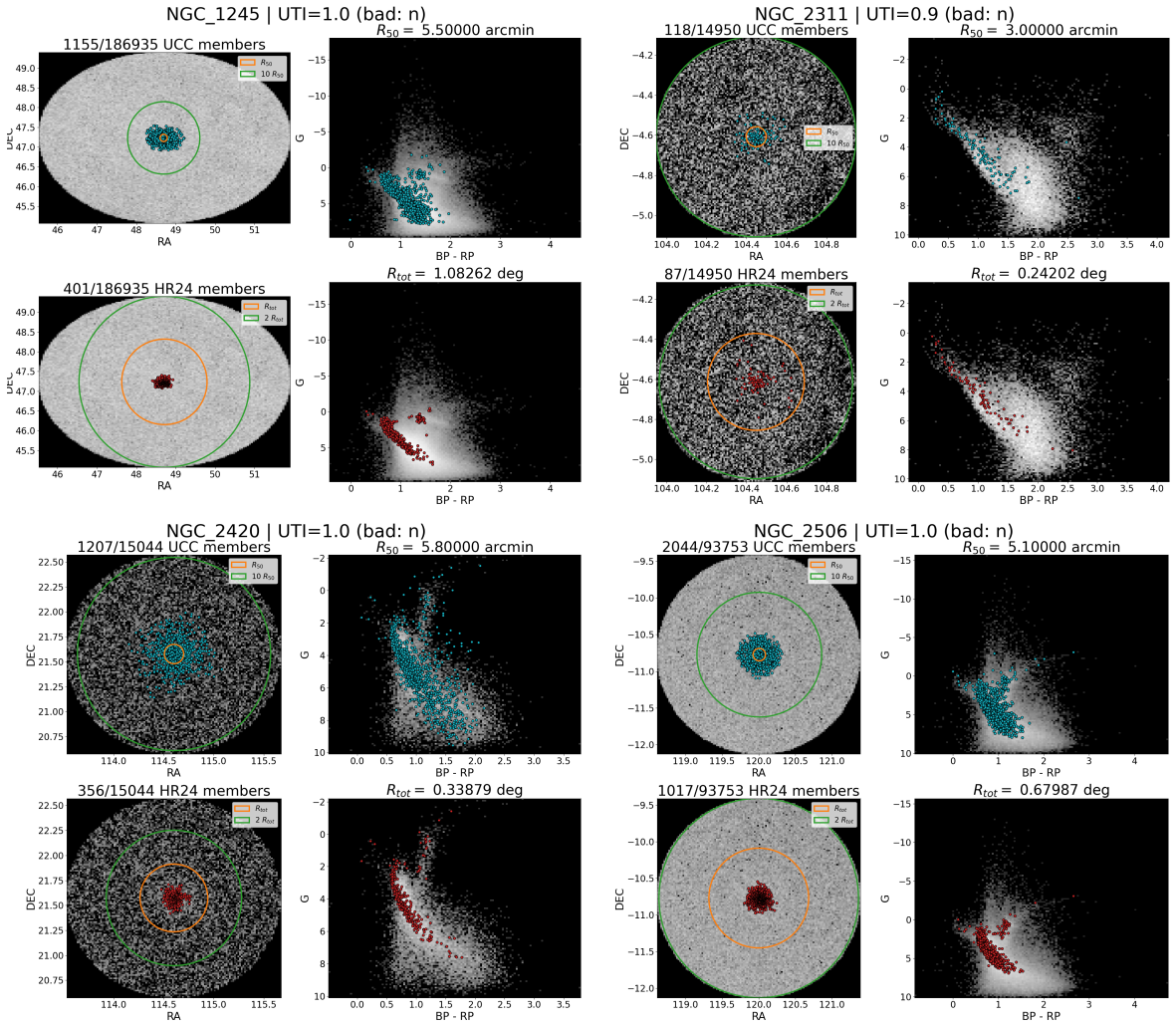


Figure 2: Gaia DR3 fields obtained using each OC's reported coordinates, with concentric rings denoting the search fields radii, along with the corresponding Gaia DR3 CMD. The upper panels of each OC overlays the reported cluster members in the UCC in blue, while the lower panels show the HR24 members overlaid in red.

quantifies the reliability of the Gaia EDR3 astrometric solution. It has been previously used with success to discriminate cluster and star formation groups from a population of field stars without performing a hard cut on either magnitude or RUWE (Hunt & Reffert, 2023; Rottensteiner et al., 2026).

3.2 HDBSCAN Clustering Pipeline

Field star discrimination is performed using HDBSCAN (Campello et al., 2013), a hierarchical density-based machine learning algorithm adept at finding clusters of data, whose hierarchical approach improves upon its predecessor DBSCAN's performance by not being biased towards a prescribed density.

HDBSCAN takes two input parameters: a minimum number of neighboring points

for a given cluster center (m_{pts}) and a minimum cluster size (m_{cls}) for a given *data cluster* (for the purposes of this paper, the term *data cluster* is used to denote a grouping of data, but not necessarily a physical stellar cluster) is considered significant and not labeled as noise in the data. The membership determination pipeline in its current state runs the HDBSCAN clustering algorithm for a grid of $10 \leq m_{cls} \leq 5000$ in steps of 20. m_{pts} was fixed to a value of 2 to reduce compute time, while the choice of $m_{pts} = 2$ was motivated to allow HDBSCAN to remain sensitive to more sparse clusters; when applied to more populated clusters such as NGC 1245, NGC 2311, NGC 2420, and NGC 2506, the hierarchical nature of HDBSCAN eventually collapses smaller groupings to a single cluster.

3.3 Membership Determination Results

The HDBSCAN data clusters are subsequently processed to find the cluster of interest, as HDBSCAN will almost always return more than one singular cluster per field. The most correct cluster is identified by its *average centroid linkage distance* (D_c) to the literature cluster:

$$D_c(H, L) = \frac{1}{|H| + |L|} \left\{ \sum_{\vec{x} \in H} d(\vec{x}, \vec{c}_L) + \sum_{\vec{y} \in L} d(\vec{y}, \vec{c}_H) \right\} \tag{1}$$

Where H is the data cluster identified from the HDBSCAN pipeline, L is the cluster composed of the members reported in the literature (taken from Hunt & Reffert, 2024 (henceforth HR24) when available, else from the UCC (Perren et al., 2023)), $\vec{x}$ and $\vec{y}$ are the normalized 5D astrometric parameters of the members in clusters H and L respectively, the function d returns the Euclidean distance between two points, and $\vec{c}_H$ and $\vec{c}_L$ are the center positions of clusters H and L respectively. The data cluster H with the lowest D_c to the literature cluster L is taken as the new solution.

Figure 3 shows the results of the HDBSCAN pipeline on each of the four clusters in this work, having accepted only the members with a membership probability greater than 50%. In 3 of the 4 clusters the pipeline retrieves both an adequate overlap with the membership lists in the HR24 catalog, while identifying a significant amount of new member stars, both in the spatial outskirt regions of each cluster as well as an appreciable locus of new stars (as is the case of NGC 2311) of faint stars. NGC 2420 presents a case where the HDBSCAN pipeline did not retrieve an adequate amount of stars when compared with the literature, and, perhaps more significantly, did not identify any new member stars, likely due to the search field size when compared to the search radius used by HR24, processing multiple Gaia DR3 HEALPix pixels at once in search of OCs. As a result, for the subsequent photometric analysis, we used the HR24 membership catalog for NGC 2420, while for the rest of the clusters we used the list obtained from the HDBSCAN pipeline.

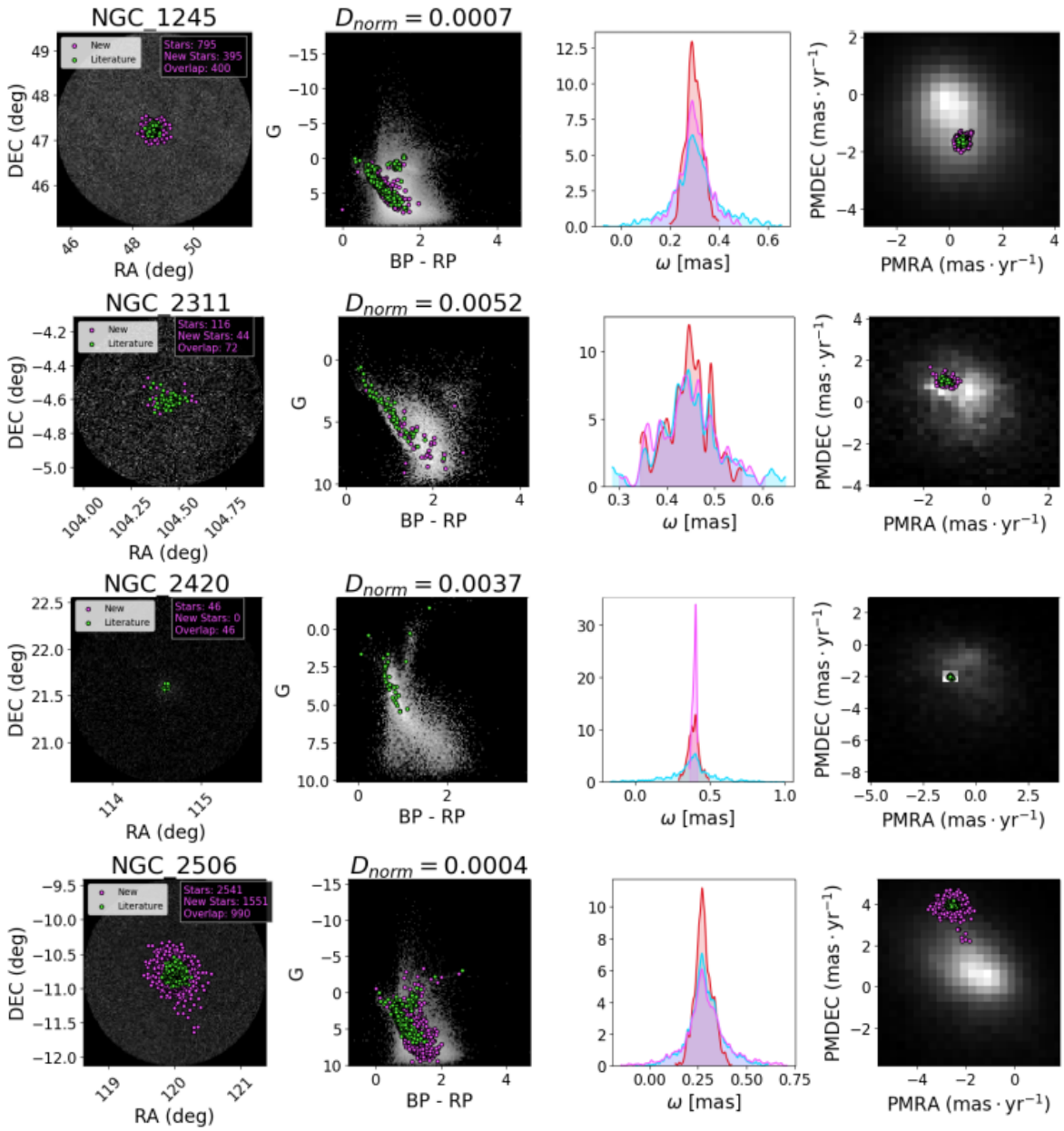


Figure 3: Matched clusters obtained by HDBSCAN for each of the four OCs in this work. Their spatial distribution can be seen in columns 1 and 3, where column 3 shows the KDE distribution of the cluster members in the UCC catalog (cyan), HR24 (red), and the recovered HDBSCAN cluster (magenta). Column 2 shows the CMD in the Gaia bands, and column 4 shows the clusters' distribution in the proper motion space. In all but one of the clusters we manage to retrieve members further than the previously reported furthest members from the core. D_{norm} represents the average centroid linkage distance (D_c) of each retrieved cluster on the scaled astrometric parameters of each member star.

4 Photometric Analysis

Using the membership list from the HDBSCAN pipeline, we performed a photometric analysis of each OC to determine their physical parameters: metallicity (Z; calculated using [Fe/H]), age ($\log t$), distance given as the distance modulus (d_m), visual extinction (A_V), and Galactic extinction curve (R_V). This was done using each cluster's computed Gaia CMDs as a first approximation and fitting PARSEC theoretical isochrones (Bressan et al., 2012), where the fit was done using two different approaches to explore the existing tooling available in the literature.

4.1 ASteCA

The *Automated Stellar Cluster Analysis* (ASteCA) package (Perren et al., 2015) is an open-source package for cluster analysis. ASteCA is organized into several modules for handling cluster membership probabilities, isochrone handling, synthetic cluster generation from theoretical isochrones, observation calibrations for comparing synthetic clusters with real observations, and likelihood calculations for comparing synthetic CMDs with a synthetic cluster's CMD. Since each clusters' stellar members were already determined from the HDBSCAN pipeline, we only made use of the `Synthetic` module to generate synthetic clusters calibrated on the Gaia member observations.

An *approximate Bayesian computation* (ABC) combined with *sequential Monte-Carlo* (ABC-SMC) approach was employed for parameter inference with ASteCA using the `pyABC` package (Schälte et al., 2022). Unlike other methods of parameter inference, ABC-SMC does not depend on the evaluation of a likelihood function, but rather only on a measurement of a distance metric of the generated model (in this case, the synthetic cluster created from a parameter vector sampled from a given target distribution) and the observed data (the observational Gaia DR3 CMD from the member stars identified from the HDBSCAN pipeline), denoted as ϵ.

In addition to the main intrinsic parameters that dictate the general shape of a theoretical isochrone, unresolved binary systems can have an impact on a cluster's CMD, appearing as an excessively red star and manifesting as a broadening of the cluster's main sequence (Li et al., 2020). When generating a synthetic cluster, ASteCA generates binary systems with the given probability ($P_b(m_1)$) as a function of the primary's mass (m_1) with free parameters α and β:

$$P_b(m_1) = \alpha + \beta \left(\frac{m_1}{1.4 + m_1} \right) \tag{2}$$

4.1.1 ABC Sampling

Parameter inference with `pyABC` was performed by sampling from a uniform distribution for all cluster intrinsic and extrinsic parameters. Parameter ranges can be seen in Table 1. Although this imposes an initial restriction on the parameter space initially explored by `pyABC`, these are not hard limits, as proposal parameter vectors carry the sampler beyond the limits defined by the prior distributions (Sisson et al., 2007).

Parameter	**Minimum**	**Maximum**
$\log t$	7.0	9.5
Z	0.005	0.0197
d_m	7.0	10.5
A_V	0.0	3.0
R_V	2.0	4.0
α	0.0	0.2
β	0.8	0.99
$E(B-V)$	0	4.0

Table 1: Prior parameter sample ranges.

The logarithmic Poisson likelihood ratio (PLR) ($\log p$) defined within ASteCA is used as the distance metric for `pyABC`:

$$\log p \approx \sum_{i}^{B} \left[\log \Gamma \left(n_i + m_i + \frac{1}{2} \right) - \log \Gamma \left(m_i + \frac{1}{2} \right) \right] \tag{3}$$

The PLR is calculated on a 2D histogram of the Gaia DR3 CMD composed of B bins, where n_i and m_i are the number of observed and synthetic stars in bin i, respectively. Each cluster was sampled until either $\epsilon \leq 0.01$, $\Delta\epsilon \leq 10^{-9}$, or 3000 iterations were reached.

4.2 `simul_fit`

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